

Sparse Factor Modeling for Equity Return Prediction Using Convex Optimization

MSML 604 – Introduction to Optimization | Project Abstract

By: Renuka Oladri, UID: 122279735

Saranya Roy, UID: 121094186

1. Problem Statement and Motivation

One of the major issues in quantitative finance is to predict stock market returns using a vast set of potential predictors. The Fama-French 5-Factor model enhanced with the Momentum factor presents an established way of explaining cross-section return variation. In reality, however, not all of the factors used will actually add value to the model, either because they are redundant, poorly correlated with returns, or noisy factors that adversely impact out-of-sample results. Using all the factors available would simply mean over-fitting on a sample without generalizing outside of it.

On the other hand, Fama-French factors themselves suffer from multicollinearity which increases the variance of the OLS estimator. For such a reason, there is a clear need for automated factor selection which would simultaneously achieve good predictive power and parsimony of the model. It can be solved by means of convex optimization methods.

2. Application Context

Factor models are an essential component of systematic investment and quantitative finance. Analysts working within asset management firms are constantly confronted by the problem of deciding what risk factors to incorporate in their predictive model frameworks to avoid underfitting or overfitting risks. This project attempts to tackle this problem in an academic environment through the use of open-source data provided by the Kenneth French Data Library.

In particular, the five factors (Market, SMB, HML, RMW, CMA) from Fama-French together with the Momentum factor will be used as predictors while the monthly returns of 25 size-and-book-to-market-ordered portfolios will be the target variables.

3. Optimization Formulation

The main point of this work is the formulation of three different regularized regression techniques as convex optimization problems. Let X be a design matrix of factor returns in $\mathbb{R}^{n \times p}$ and y be a response vector of portfolio returns in $\mathbb{R}^n$, then we can consider the following convex optimization problems:

Ridge Regression: minimize $\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2$

LASSO: minimize $\|y - X\beta\|_2^2 + \lambda \|\beta\|_1$

Elastic Net: minimize $\|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$

Convexity holds for each of these criteria, since the function being squared is strictly convex, while both norms are convex functions, and hence their sum retains convexity. For each case, KKT optimality conditions, as well as the dual problem, where relevant, will be obtained. The complementary slackness conditions associated with the LASSO will be examined to explain the reason behind obtaining zero coefficients using L1 penalty on the basis of theory.

4. Proposed Methodology

Algorithmically, we will implement two techniques for solving the problem from scratch: proximal gradient descent for LASSO and Elastic Net (using the proximal operator of L1 norm, which is also known as soft thresholding), as well as coordinate descent technique to compare. We will use CVXPY, a package for disciplined convex programming, as a competitor and will compare convergence speed, accuracy, and efficiency of our methods to this package.

In practice, we will test performance of our models based on the walk forward backtesting scheme applied to the set of 25 Fama-French portfolios. At each step, the model is trained on some history and predicted out-of-sample, which reflects practical application of the model to make predictions. Metrics will include OOS R2, out-of-sample mean squared prediction error, and the Sharpe ratio of the long-short portfolio based on predictions.

5. Goals and Expected Outcomes

In the first place, we seek to mathematically define LASSO, Ridge, and Elastic Net and analyze their properties as convex programs, such as KKT conditions, duality and behavior. In addition, we seek to develop our own algorithms for solving them and investigate their convergence behavior experimentally by means of visualizing their regularization paths and loss functions. Finally, we plan on testing the predictive ability of these models through out-of-sample evaluation in a financial forecasting task.

We posit that LASSO and Elastic Net will be capable of providing sparser and more consistent factor selection compared to Ridge. Additionally, we posit that LASSO and Elastic Net will have better out-of-sample generalization as a result of factor selection due to the introduction of an L1 term. On the other hand, Ridge will excel when there is strong multicollinearity in factors, since all terms are regularized. Nevertheless, Ridge will retain all factors while Elastic Net will select them.

All-in-all, the objective of this project is to show that convex optimization algorithms are much more than mere computational tricks, as they can offer sound model selection solutions in quantitative finance.